1 **(a)** $\log_{10} x$ OR $\log x$ **[B1]**

(b) -2 **[B1]**

(c) $\log_{10} 45$ OR $\log 45$ **[B1]**

[Total Marks: 3]

2 **(a)** $\log 1000 = 3$ **[B1]**

(b) 0.01 or $\frac{1}{100}$ **[B1]**

(c) $\frac{\log 12}{\log 5}$ **[B1]**

[Total Marks: 3]

3 **(a)** 2 **[B1]**

(b) -3 **[B1]**

(c) x **[B1]**

[Total Marks: 3]

4 **(a)** 4 **[B1]**

(b) $\log_5 125 = 3$ **[B1]**

(c) 64 **[B1]**

[Total Marks: 3]

5 **(a)** 32 **[B1]**

(b) -2 **[B1]**

(c) $t = \log_k P$ **[B1]**

[Total Marks: 3]

6 **(a)** $\log_5 125 = 3$ **[B1]**

(b) 3 **[B1]**

(c) $x = 2^5$ **[M1]**

32 **[A1]**

[Total Marks: 4]

7 **(a)** $\log 100000 = 5$ **[B1]**

(b) 10^3 **[M1]**

1000 **[A1]**

(c) $\log x$ **[B1]**

[Total Marks: 4]

8 **(a)** **(i)** 3

3 **[B1]**

(ii) 5

5 **[B1]**

(b)

=

3

4

=

81

x=3

4

=81 **[B1]**

(c) log

logx (or

log

10

log

10

x) **[B1]**

[Total Marks: 4]

9 **(a)** **(i)** 5 **[B1]**

(ii) 0.01 or $\frac{1}{100}$ **[B1]**

(iii) 6 **[B1]**

(b) $\log x$ or $\log_{10} x$ **[B1]**

[Total Marks: 4]

10 **(a)** 3 **[B1]**

(b) 10^{-2} or $\frac{1}{100}$ **[M1]**

0.01 **[A1]**

(c) $\log_5 125$ **[B1]**

[Total Marks: 4]

11 **(a)** $\log_{10} 10000 = 4$ (or $\log 10000 = 4$) **[B1]**

(b) Uses index form $x = 5^3$ **[M1]**

125 **[A1]**

(c) $a = 7$ **[B1]**

$b = 25$ **[B1]**

[Total Marks: 5]

12 **(a)** 4 **[B1]**

(b) Writes $\frac{1}{9} = 3^{-2}$ **[M1]**

-2 **[A1]**

(c) Writes $y = \log_3 x$ as $x = 3^y$ (or $y = 3^x$) **[M1]**

3^x **[A1]**

[Total Marks: 5]

13 **(a)** **(a)** $p(1) + q = 8$ or $p(2) + q = 11$ **[B1]**

A valid method to solve simultaneous equations e.g. $(2p + q) - (p + q) = 11 - 8$ **[M1]**

$p = 3$, $q = 5$ **[A1]**

(b) $3 \log x + 5 = 23 \implies 3 \log x = 18 \implies \log x = 6$ **[M1]**

$1\,000\,000$ or 10^6 **[A1]**

(c) $x = 3 \log y + 5 \implies \log y = \dfrac{x - 5}{3}$ **[M1]**

$10^{\frac{x-5}{3}}$ **[A1]**

[Total Marks: 7]

14 **(a)** **(a)** $\log_6 y$ **[B1]**

(b) $x = \dfrac{\log 20}{\log 3}$ or $x \log 3 = \log 20$ **[M1]**

2.73 **[A1]**

[Total Marks: 3]

15 **(a)** 250×1.08^{10} **[M1]**

540 **[A1]**

(b) $250 \times 1.08^t = 1000$ **[M1]**

$1.08^t = 4 \implies t = \dfrac{\log 4}{\log 1.08}$ **[M1]**

18.0 **[A1]**

[Total Marks: 5]

16 **(a)** 15000×0.8^4 **[M1]**

6144 **[A1]**

(b) $7500 = 15000 \times 0.8^t$ **[M1]**

$0.5 = 0.8^t \implies t = \dfrac{\log 0.5}{\log 0.8}$ **[M1]**

3.11 **[A1]**

[Total Marks: 5]

17 **(a)** $25000 \times (1 - 0.12)^4$ or 25000×0.88^4 **[M1]**

14992 **[A1]**

(b) $25000 \times 0.88^n = 10000$ or $0.88^n = 0.4$ **[M1]**

$n = \dfrac{\log 0.4}{\log 0.88}$ **[M1]**

$n = 7.16...$ **[A1]**

8 **[B1]**

[Total Marks: 5]

18 **(a)** $2x = \dfrac{\log 50}{\log 3}$ **[M1]**

$2x = 3.56\ldots$ **[A1]**

1.78 **[A1]**

(b) $\log y = x - 4$ **[M1]**

$\log y + 4$ **[A1]**

[Total Marks: 5]

19 **(a)** **(a)** 5000 **[B1]**

(b) $5000 \times 1.12^t = 20000$ **[M1]**

$1.12^t = 4$ **[M1]**

$t = \dfrac{\log 4}{\log 1.12}$ **[M1]**

12.2 **[A1]**

[Total Marks: 5]

20 **(a)** 8000×1.035^n **[B1]**

(b) Sets up the inequality or equation: $8000 \times 1.035^n > 12000$ OR $1.035^n = 1.5$ **[M1]**

Uses logarithms to solve for n: $n = \dfrac{\log_{10} 1.5}{\log_{10} 1.035}$ **[M1]**

12 **[A1]**

(c) $x = \dfrac{\log_{10} 55}{\log_{10} 2}$ **[M1]**

5.78 **[A1]**

[Total Marks: 6]

21 (a) $x \log 3 = \log 20$ **[M1]**

$x = \dfrac{\log 20}{\log 3}$ **[M1]**

2.73 **[A1]**

(b) $100 = 500 \times 0.85^t \implies 0.85^t = 0.2$ **[M1]**

$t \log 0.85 = \log 0.2 \implies t = \dfrac{\log 0.2}{\log 0.85}$ **[M1]**

9.90 **[A1]**

[Total Marks: 6]

22 (a) 4500×1.08^5 **[M1]**

6612 **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ oe **[M1]**

$t = \dfrac{\log(20/9)}{\log 1.08}$ **[M1]**

10.38 **[A1]**

[Total Marks: 6]

23 (a) 4500×1.08^5 **[M1]**

6611.98 (or 6610 to 3 significant figures) **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ or $1.08^t = \dfrac{20}{9}$ **[M1]**

$t = \dfrac{\log\left(\dfrac{20}{9}\right)}{\log 1.08}$ **[M1]**

10.4 (10.376...) **[A1]**

[Total Marks: 6]

24 **(a)** 0.78
0.78 (accept
0.75
0.75 to
0.80
0.80) **[B1]**

(b) 3.2
3.2 (accept
3.1
3.1 to
3.3
3.3) **[B1]**

(c) (
+
1
)
log
3
=
log
20
(x+1)log3=log20 or
+
1
=
log
20
log
3
x+1=
log3
log20
[M1]

+
1
=
2.7268
...
x+1=2.7268... **[M1]**

1.73
1.73 **[A1]**

(d) 2
3
=
10
1.4
2x3=10
1.4
(or
2
3

=
25.118
...
2x3=25.118...) **[M1]**

14.1
14.1 **[A1]**

[Total Marks: 7]

25 **(a)** Substitutes $t = 5$ into the formula: 400×0.88^5 **[M1]**

211.1 **[A1]**

(b) Sets up the equation $400 \times 0.88^t = 200$ and simplifies to $0.88^t = 0.5$ **[M1]**

$t = \dfrac{\log 0.5}{\log 0.88}$ **[M1]**

5.42 **[A1]**

(c) Sets up the equation $800 \times k^{10} = 300$ **[M1]**

$k^{10} = 0.375$ **[M1]**

0.907 **[A1]**

[Total Marks: 8]

26 **(a)** 8000×1.045^5 **[M1]**

9969.46 (accept 9969 or 9970) **[A1]**

(b) $1.045^t = \dfrac{15000}{8000}$ or 1.875 **[M1]**

$t = \dfrac{\log 1.875}{\log 1.045}$ or 14.28... **[M1]**

15 **[A1]**

(c) $\log(7^{x-1}) = \log 250$ or $x - 1 = \log_7 250$ **[M1]**

$x - 1 = \dfrac{\log 250}{\log 7}$ or $x - 1 = 2.837...$ **[M1]**

3.84 **[A1]**

[Total Marks: 8]

27 **(a)** 12800 **[B1]**

(b) $800 \times 2^t = 102400 \implies 2^t = 128$ **[M1]**

$t = 7$ **[A1]**

(c) $1500 \times 3^t = 20000 \implies 3^t = \dfrac{40}{3}$ (or 13.33...) **[M1]**

Uses logarithms to solve $3^t = \dfrac{40}{3}$, e.g. $t = \dfrac{\log(40/3)}{\log 3}$ **[M1]**

2.36 (from 2.358...) **[A1]**

(d) $2000 \times 1.05^n = 4000 \implies 1.05^n = 2$ **[M1]**

$n = \dfrac{\log 2}{\log 1.05}$ (or 14.2...) **[M1]**

15 **[A1]**

[Total Marks: 9]

28 (a) $y = 4 \times 10^{2x} - 7 \implies y + 7 = 4 \times 10^{2x}$ **[M1]**

$\dfrac{y+7}{4} = 10^{2x} \implies 2x = \log\left(\dfrac{y+7}{4}\right)$ **[M1]**

$\dfrac{1}{2}\log\left(\dfrac{x+7}{4}\right)$ **[A1]**

(b) $4 \times 10^{2x} - 7 = 53 \implies 4 \times 10^{2x} = 60$ **[M1]**

$10^{2x} = 15 \implies 2x = \log 15$ **[M1]**

0.588 **[A1]**

[Total Marks: 6]

29 (a) Exponential curve passing through $(0, 1)$ and $(1, 2)$ **[B1]**

Curve passing through $(3, 8)$ and approaching the negative x-axis as an asymptote **[B1]**

(b) $\dfrac{\log x}{\log 2}$ (or $\log_2 x$) **[B1]**

(c) Logarithmic curve passing through $(1, 0)$ and $(2, 1)$ **[B1]**

Curve passing through $(8, 3)$ and approaching the negative y-axis as an asymptote **[B1]**

(d) $x = \dfrac{\log 12}{\log 2}$ **[M1]**

3.58 **[A1]**

[Total Marks: 7]

30 (a) (i) 250 **[B1]**

(ii) $250 \times 0.92^t = 50$ or $0.92^t = 0.2$ **[M1]**

$t = \dfrac{\log 0.2}{\log 0.92}$ **[M1]**

19.3 **[A1]**

(b) $800 \times 1.06^n > 2000$ or $1.06^n = 2.5$ **[M1]**

$n = \dfrac{\log 2.5}{\log 1.06}$ $(= 15.7\ldots)$ **[M1]**

16 **[A1]**

[Total Marks: 7]

31 **(a)** x-intercept $= (1, 0)$ **[B1]**

Asymptote: $x = 0$ **[B1]**

(b) 5.0 (accept 4.9 to 5.1) **[B1]**

(c) Takes logarithms of both sides: $(2x - 1)\log 3 = (x + 2)\log 5$ **[M1]**

Expands and collects x terms: $x(2\log 3 - \log 5) = 2\log 5 + \log 3$ **[M1]**

$x = \dfrac{2\log 5 + \log 3}{2\log 3 - \log 5}$ (or $\dfrac{\log 75}{\log 1.8}$) **[M1]**

7.34 (7.3414...) **[A1]**

[Total Marks: 7]

32 **(a)** Substitutes given values into the formula: $250 = 800 \times 10^{-15k}$ **[M1]**

Rearranges to isolate the exponential: $10^{-15k} = 0.3125$ **[M1]**

Takes logarithms to solve for k: $-15k = \log_{10}(0.3125) \implies k = \dfrac{\log_{10}(0.3125)}{-15} = 0.03367...$ **[A1]**

(b) Substitutes $m = 50$ and $k = 0.03367...$ into the formula: $50 = 800 \times 10^{-0.03367t}$ **[M1]**

Isolates the exponential and takes logarithms: $-0.03367t = \log_{10}\left(\dfrac{50}{800}\right)$ **[M1]**

35.8 (accept 35.7 to 35.8) **[A1]**

(c) Rearranges to $10^{-kt} = \dfrac{m}{800}$ and converts to logarithmic form: $-kt = \log_{10}\left(\dfrac{m}{800}\right)$ **[M1]**

$t = -\dfrac{1}{k}\log_{10}\left(\dfrac{m}{800}\right)$ OR $t = \dfrac{1}{k}\log_{10}\left(\dfrac{800}{m}\right)$ **[A1]**

[Total Marks: 8]

33 **(a)** $g(100) = \log 100 = 2$ **[M1]**

$f(2) = 5 \times 10^2 = 500$ **[A1]**

(b) $y = 5 \times 10^x$ leads to $10^x = \dfrac{y}{5}$ **[M1]**

$\log\left(\dfrac{x}{5}\right)$ or $\log x - \log 5$ **[A1]**

(c) $5 \times 10^x = 200$ leads to $10^x = 40$ **[M1]**

$x = \log 40 = 1.602...$ **[A1]**

1.60 **[B1]**

(d) $g(f(x)) = \log(5 \times 10^x)$ **[M1]**

$= \log 5 + \log 10^x = \log 5 + x$ **[M1]**

$x + \log 5$ **[A1]**

[Total Marks: 8]

34 **(a)** $5 \times 3^{-2} - 2$ **[M1]**

$-\dfrac{13}{9}$ **[A1]**

(b) $y = 5 \times 3^x - 2 \implies y + 2 = 5 \times 3^x$ **[M1]**

$3^x = \dfrac{y+2}{5}$ **[M1]**

$\dfrac{\log\left(\frac{x+2}{5}\right)}{\log 3}$ **[A1]**

(c) $5 \times 3^x - 2 = 500 \implies 3^x = 100.4$ **[M1]**

$x = \dfrac{\log 100.4}{\log 3}$ **[M1]**

4.20 **[A1]**

[Total Marks: 8]

35 **(a)** $3 = \log_a 8$ **[M1]**

$a^3 = 8 \implies a = 2$ **[A1]**

(b) **(i)** Smooth exponential curve passing through $(0, 1)$ **[B1]**

Correct shape passing through $(1, 2)$, $(2, 4)$ and $(3, 8)$ **[B1]**

(ii) $y = 0$ (accept the x-axis) **[B1]**

(c) Rearranges to $\log_2 x = 8 - x$ **[M1]**

Draws the line $y = 8 - x$ correctly on the grid (passing through $(0, 8)$ and $(8, 0)$) **[M1]**

$x = 5.5$ (accept 5.4 to 5.6) **[A1]**

[Total Marks: 8]

36 **(a)** Substitutes $x = 3$: $4 \times 10^{1.5}$ **[M1]**

126 **[A1]**

(b) Sets $y = 4 \times 10^{0.5x}$ and rearranges to $\dfrac{y}{4} = 10^{0.5x}$ **[M1]**

$\log\left(\dfrac{y}{4}\right) = 0.5x$ **[M1]**

$2\log\left(\dfrac{x}{4}\right)$ **[A1]**

(c) $4 \times 10^{0.5x} = 200$

$10^{0.5x} = 50$ [M1]

$0.5x = \log 50$ **[M1]**

3.40 **[A1]**

[Total Marks: 8]

37 **(a)** $a \times b^0 = 12 \implies a = 12$ **[B1]**

$12 \times b^3 = 96 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets up $y = 12 \times 2^x \implies 2^x = \dfrac{y}{12}$ **[M1]**

$x \log 2 = \log\left(\dfrac{y}{12}\right)$ **[M1]**

$f^{-1}(x) = \dfrac{\log(x/12)}{\log 2}$ **[A1]**

(c) $12 \times 2^x = 500 \implies 2^x = \dfrac{125}{3}$ (41.666...) **[M1]**

$x = \dfrac{\log(125/3)}{\log 2}$ **[M1]**

5.38 **[A1]**

[Total Marks: 9]

38 (a)
24000
×
0.85
4
24000×0.85
4
(
=
12528.15
=12528.15) **[M1]**

12528
12528 **[A1]**

(b) $24000 \times 0.85^t = 9000$ or $0.85^t = 0.375$ **[M1]**

$t = \dfrac{\log 0.375}{\log 0.85}$ **[M1]**

6.04 **[A1]**

(c) $1.068^n = 2$ (or $1.068^n > 2$) **[M1]**

$n = \dfrac{\log 2}{\log 1.068} = 10.536\ldots$ **[M1]**

11 **[A1]**

(d) $2^y = \dfrac{x-3}{5}$ **[M1]**

$y = \log_2\left(\dfrac{x-3}{5}\right)$ or $y \log 2 = \log\left(\dfrac{x-3}{5}\right)$ **[M1]**

$log_2(\dfrac{x-3}{5}$ or
log
(
3
5
)

log
2
log2
log(
5
x3
)

[A1]

[Total Marks: 11]

39 **(a)** Substitutes $(0, 7)$ to find A: $7 = A \times b^0 \implies A = 7$ **[B1]**

Substitutes $(3, 56)$ with $A = 7$: $7 \times b^3 = 56 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets $y = 7 \times 2^x$ **[M1]**

Rearranges to make 2^x the subject: $2^x = \dfrac{y}{7}$ **[M1]**

Converts to logarithmic form: $\log_2\left(\dfrac{x}{7}\right)$ (or equivalent correct form) **[A1]**

(c) **(i)** $\dfrac{600}{7}$ **[B1]**

(ii) $x = \dfrac{\log(600/7)}{\log 2}$ **[M1]**

6.421 (from 6.4214...) **[A1]**

(d) **(i)** Sets $4 = \log_c 81 \implies c^4 = 81$ **[M1]**

$c = 3$ **[A1]**

(ii) $\dfrac{1}{9}$ (or 3^{-2}) **[B1]**

[Total Marks: 12]